

The Remarkable Efficiency of the Probabilistic Serial Mechanism

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Is the choice of allocation mechanism irrelevant in the random assignment problem? Existing results conflict: unlike Probabilistic Serial (PS), Random Priority (RP) is almost never ordinally efficient in markets where each object has only one copy, yet the two mechanisms are asymptotically equivalent in replica economies as the number of copies goes to infinity. Whether the one-copy divergence is quantitatively meaningful, and what happens when the number of copies grows slowly, were open questions that we resolve.

With q copies of m object types and mq agents with i.i.d. uniform preferences, we show that PS produces a near-constant average rank with high probability, and its expected average rank is at most $\log^* m + 3.4$ for every q . For fixed q , this bound is asymptotically tight. Here $\log^*$ is the iterated logarithm, a function which grows so slowly that it is below 5 whenever m is smaller than the number of atoms in the universe. On the other hand, RP's expected average rank is heavily dependent on q . When q is constant, it grows logarithmically, diverging from PS. With increasing q and m , the gap between PS's and RP's expected average ranks vanishes if and only if q grows faster than $\log m$.

PS's near-constant efficiency is robust and extends to imbalanced and correlated markets.

Keywords: probabilistic serial, random priority, random markets.

1 Introduction

The Random Priority mechanism (RP) assigns objects by drawing an order of agents and letting each choose their favourite remaining object. It is a commonly used mechanism, partly because it is strategy-proof and ex-post efficient. Yet, the probability distribution that it generates over assignments need not be ordinally efficient, meaning that agents may want to trade different probability shares over objects. To address RP's ordinal inefficiency, [Bogomolnaia and Moulin \[2001\]](#) proposed a new mechanism called Probabilistic Serial (PS). Under PS, all agents simultaneously consume their favourite available objects at the same speed. The resulting fractional assignment is not only ordinally efficient, but also envy-free, at the cost of losing strategy-proofness.

After Bogomolnaia and Moulin's seminal paper, the literature has tackled the natural follow-up question: does PS's ordinal efficiency guarantee meaningfully better assignments than RP? Two articles give sharply different answers. [Manea \[2009\]](#) showed that, in the original setup with the same number of agents and objects, the fraction of preference profiles for which RP is ordinally efficient vanishes, and thus PS and RP differ with high probability. On the other hand, [Che and Kojima \[2010\]](#) consider replica economies with a fixed set of object types. They show that as the number of object copies grows, the allocations generated by RP and PS are asymptotically equivalent. Thus, they argue, *"it may be misleading to simply examine whether a mechanism suffers ordinal inefficiencies; even if a mechanism is ordinally inefficient, the magnitude of the inefficiency may be very small"*.

At the same time, Che and Kojima note that RP's ordinal inefficiency need not vanish if the number of copies is fixed (even if arbitrarily large), and conclude that the comparison between PS and RP in finite economies remains an open question. Manea similarly notes that the Che and Kojima equivalence *"applies only to environments in which the quota of each object is considerably larger than the number of object types"*, and writes that no bound for the expected inefficiency as a function of the quotas is known. In summary, three questions remained open: i) how large is RP's inefficiency in the original setting with no replicas, ii) how does each mechanism's expected inefficiency depend on the number of copies in finite economies, and iii) how many replicas do we need to close their efficiency gap?

We answer the aforementioned three questions in this paper, assuming in the main text that agents’ preferences are drawn independently and uniformly at random.¹ We establish explicit bounds on PS’s expected average rank for every finite market size, showing that PS’s expected average rank admits a nearly constant upper bound that does not depend on the number of object replicas. Furthermore, we identify how many object copies are needed to close the efficiency gap between the two mechanisms.

To be precise, we focus on random markets with m object types with q copies each, and mq agents with preferences drawn independently and uniformly at random. We measure efficiency by the average rank generated by each mechanism under truthful preference revelation, with lower ranks meaning better allocations.² Average ranks are widely used as welfare statistics, as they measure how good the objects agents receive are, with each object’s rank weighted by its assignment probability, based purely on ordinal information without any assumptions on preference intensity. **Theorem 1** shows that PS achieves near-constant average rank, both in expectation and with high probability, regardless of the number of object copies in the economy. Its expected average rank is at most $\log^* m + 3.4$, where $\log^*$ is the number of times the natural logarithm must be applied before the result is at most one. The iterated logarithm grows extraordinarily slowly: for example, $\log^* m = 4$ even when $m = 10^{80}$, the estimated number of atoms in the universe [Cormen et al., 1990]. In the no-replicas setting ($q = 1$), RP’s expected average rank grows logarithmically, which means the gap between the mechanisms diverges. For example, with 10,000 objects with no replicas, PS’s expected average rank is at most 6.4, whereas RP’s is about 8.8 (see **Figure 1** for an illustration). This rate is sharp: for every fixed q , PS’s expected average rank is asymptotically equivalent to $\log^* m$, so it diverges at the iterated-logarithmic rate.

Theorem 2 then identifies exactly when object replication closes the PS-RP efficiency gap. The gap remains if the number of copies is fixed. As the number of object types grows, RP catches up with PS in expected average rank if and only if copies per type grow faster than the logarithm of the number of types. When

¹Appendix C.3 extends the results allowing for a class of correlated preferences drawn according to objects’ popularity scores.

²In our i.i.d. setup, PS is ordinal Bayesian incentive compatible [Dasgupta and Mishra, 2022].

copies grow more slowly than this logarithm, the gap diverges. When they grow proportionally to it, RP’s expected rank becomes bounded, but a positive gap from PS persists. [Theorem 2](#) provides a resolution to the tension in the literature. In the one-copy setting studied by Manea, the rank gap diverges. With the fixed set of types considered by Che and Kojima, increasing the number of copies makes RP catch up. Convergence comes from RP improving as larger capacities make demand fluctuations small relative to supply.

The intuition for PS’s upper bound is as follows. When few object types remain available, agents may have to consume goods they rank poorly. PS performs well because this part of the process is short. All agents continue eating until time one, so the surviving types must serve the entire population. We show that most of these types have many consumers and therefore exhaust their remaining supply quickly.

The challenge is to establish this demand bound for the types that actually survive, since their survival depends on agents’ earlier choices. We prove that the bound holds throughout the process with high probability, and use it to divide consumption into an iterated-logarithmic number of phases, each contributing at most a constant to average rank.

The matching lower bound has a simple source. Types that no agent has yet chosen retain their entire supply, and before the process can reach a much smaller set of surviving types, most of that supply must be consumed. Each such episode forces a contribution of nearly one unit to expected rank. For fixed q , the episodes occur at scales related by repeated exponentiation, yielding an iterated-logarithmic number of them.

Further Literature. After Bogomolnaia and Moulin, much has been written about PS and its properties. Yet, only a handful of papers have studied the quality of assignments that PS generates in practice. [Hugh-Jones et al. \[2014\]](#) conduct an experiment comparing PS to RP in the lab in small markets with only 3 agents and 3 objects. They find evidence of PS being more efficient than RP under observed non-truthful behaviour in one of two relevant treatments. For markets where each object has many copies, [Pathak \[2006\]](#) compares PS with RP using submitted preferences

from over eight thousand students in the 2003-04 supplementary round of New York City’s high school match. The PS lottery stochastically dominates the RP lottery for 50% of students, while the reverse holds for 6%. Nevertheless, the differences in assignment probabilities are small for most students. Kornbluth and Kushnir [2025] study a course allocation problem with roughly six thousand students and seven hundred fifty courses and find that, among students whose schedules change, PS increases mean estimated utility by 31.8% for first-year students, while reducing it by 3.3%, 16.9%, and 8.6% for second-, third-, and fourth-year students, respectively. In a recent paper, Eraslan et al. [2026] compare instead PS to RP when assigning roughly nine hundred students to high schools in Seattle. They also estimate cardinal preferences over schools, and with the estimated utilities find that both mechanisms generate almost identical average normalized expected utility.

With regards to the incentive properties of PS, there is a wide theoretical literature with mixed conclusions. The closest paper to us in this literature is Dasgupta and Mishra [2022], who show that if all agents have the uniform prior (as in the i.i.d. uniform model), PS is ordinal Bayesian incentive compatible. Kojima and Manea [2010] is also related to our work, showing that truthful preference reporting becomes a weakly dominant strategy in large replica markets. On the negative side, PS may yield equilibrium outcomes that are neither ordinally efficient nor envy free in specific markets [Ekici and Kesten, 2016]. Yet, computing a best response in PS is computationally complex, and can only yield a modest utility gain versus truthful reporting [Aziz et al., 2015, Huang et al., 2024]. These results are helpful to explain why most subjects who manipulate PS in practice end up with lower payoffs [Hugh-Jones et al., 2014]. In their experiment, roughly 20% and 27% of reports are insincere under RP and PS, respectively, and misreporting is significantly more frequent under PS than under RP in only one of the eight decision situations.

Random Markets. There is an extensive literature analyzing the performance of resource allocation algorithms on average rather than on specific instances. This literature, starting with seminal contributions from computer scientists to settle the performance of the Gale-Shapley deferred acceptance procedure in one-to-one markets [Knuth, 1976, Wilson, 1972], has been extended to analyze the performance of a wider family of allocation mechanisms under a variety of market conditions

[Ashlagi et al., 2017, Che and Tercieux, 2017, Frieze and Pittel, 1995, Knuth, 1996, Ortega and Klein, 2023, Ortega et al., 2026, Pittel, 1989, 1992, Pritchard and Wilson, 2023].

A recurring theme in this literature is that different mechanisms become equivalent as markets grow large. For example, Che and Tercieux [2018] establish asymptotic payoff equivalence among Pareto-efficient mechanisms under random preferences, and show that their *normalized* rank distributions also converge.³ Relatedly, Liu and Pycia [2016] establish asymptotic allocation equivalence among regular mechanisms that are symmetric, asymptotically efficient, and asymptotically strategy-proof, while keeping the set of object types fixed. Our results characterize the comparison between PS and RP when the number of types also grows. We identify exactly when their unnormalized expected average ranks converge: copies per type must grow faster than the logarithm of the number of types.

Outline. Section 2 introduces the model and states the rank bounds for PS. Section 3 compares PS with RP and states the quota threshold for convergence. Section 4 presents the upper and lower bound arguments for Theorem 1 and the refinement giving its explicit expectation bound. Section 5 presents the extensions to unequal quotas, unbalanced markets and correlated preferences. Appendix A contains the supporting PS estimates, Appendix B proves the RP comparison, and Appendix C proves the extensions.

2 Model and PS's Average Rank

We use I and G to denote sets of n agents and m types of objects, respectively.⁴ Each type has $q \geq 1$ identical copies and $n = mq$. Each agent demands one unit of shares of potentially different types. Agent i has a strict preference order $\succ_i$ over the set of object types G . A preference profile is denoted by $\succ = (\succ_i)_{i \in I}$. A market is a tuple $(I, G, q, \succ)$.

³In the one-copy setting, both PS and RP average ranks divided by market size converge to zero, even though the gap between them diverges.

⁴We use the following asymptotic notation. For positive sequences a_n and b_n , we write $a_n = O(b_n)$ if there exists a constant $C < \infty$ such that $a_n \leq Cb_n$ for all sufficiently large n , and $a_n = o(b_n)$ if $a_n/b_n \rightarrow 0$. Subscripts on O and Θ indicate the fixed parameters on which the constants may depend. Θ denotes bounds above and below by positive constant multiples.

For each object type $g \in G$, let

$$\text{rank}_i(g) := 1 + |\{h \in G : h \succ_i g\}| \quad (1)$$

denote the position of object type g in agent i 's preference list, so that rank 1 is the most preferred object type, and rank m is the worst. Every agent has an independently drawn preference, uniform over the $m!$ strict preference orders over G .

A random assignment is a non-negative $n \times m$ matrix

$$x = (x_{ig})_{i \in I, g \in G}, \quad (2)$$

where x_{ig} is the probability share of good g assigned to agent i . Each row sums to one and each column sums to q : every agent receives one unit in total, and the full supply of every type is allocated.

A mechanism M maps each preference profile into a random assignment. Under Probabilistic Serial (PS), all agents eat continuously at unit speed. At each time, every agent eats her favourite object with positive remaining supply. When an object is exhausted, each of its current consumers immediately moves to her favourite surviving good. Under Random Priority (RP), a uniform order of agents is drawn, and each agent chooses one copy of her favourite object type with positive remaining supply.

For $M \in \{\text{PS}, \text{RP}\}$, let

$$M(\succ) = (M_{ig}(\succ))_{i \in I, g \in G} \quad (3)$$

denote the assignment produced by mechanism M at profile $\succ$.

For a fixed preference profile, the average rank of the assignment produced by mechanism M is

$$\text{Rank}(M; \succ) := \frac{1}{n} \sum_{i \in I} \sum_{g \in G} \text{rank}_i(g) M_{ig}(\succ). \quad (4)$$

Let $\succ$ denote the random preference profile. We write

$$\text{Rank}(M) := \text{Rank}(M; \succ) \quad (5)$$

for the resulting random variable. Expectations and probabilities are taken over $\succ$. For RP, $\text{RP}(\succ)$ already averages over the random priority order, so the only remaining randomness in $\text{Rank}(\text{RP})$ comes from preferences. These quantities depend on m and q , with $n = mq$. We suppress these subscripts throughout to lighten notation.

Let $\log^* x$ denote the iterated logarithm of x , i.e. the least integer $k \geq 0$ such that applying the natural logarithm k times to x yields a value at most one. [Theorem 1](#) gives an explicit bound on PS's expected average rank, an almost-constant bound with high probability, and the asymptotic growth rate of expected rank for every fixed quota.⁵

THEOREM 1. *For all positive integers m and q , with $n = mq$,*

$$\mathbb{E} \text{Rank}(\text{PS}) \leq \log^* m + 3.4. \quad (6)$$

Moreover, the same iterated-logarithmic order holds with high probability: there are constants $C, c > 0$ such that:

$$\mathbb{P}(\text{Rank}(\text{PS}) > C(1 + \log^* m)) \leq Cn^{-3} + e^{-cn}. \quad (7)$$

For every fixed q ,

$$\mathbb{E} \text{Rank}(\text{PS}) \sim \log^* m \quad \text{as } m \rightarrow \infty. \quad (8)$$

The upper and lower bounds are proved in [Section 4](#), with auxiliary estimates in [Appendix A](#).

3 Comparison with Random Priority

The following lemma describes how RP's expected average rank depends on the number of types and their quotas.

LEMMA 1. *For all integers $m \geq 2$ and $q \geq 1$,*

$$\mathbb{E} \text{Rank}(\text{RP}) = \Theta\left(1 + \frac{\log m}{q}\right), \quad (9)$$

⁵All logarithms are natural and e denotes Euler's number.

with constants independent of m and q . For $q = 1$ (so $n = m$), [Knuth \[1996\]](#) shows

$$\mathbb{E} \text{Rank}(\text{RP}) = \frac{m+1}{m}(H_{m+1} - 1) = \log m + O(1), \quad (10)$$

where $H_k := \sum_{r=1}^k 1/r$ is the k -th Harmonic number.

The general statement follows from the two-sided bounds in (96), which combine the lower bound of [Nikzad \[2022, Proposition 5.2\]](#) with the upper bound we prove in [Appendix B.1](#).

For $q = 1$, (6) and [Lemma 1](#) imply, as $m \rightarrow \infty$,

$$\mathbb{E} \text{Rank}(\text{RP}) - \mathbb{E} \text{Rank}(\text{PS}) \longrightarrow \infty. \quad (11)$$

With $q = 1$, (6) and (10) imply that PS has lower expected average rank than RP for every $m \geq 911$.⁶ With $q = 1$ and $m = n = 10,000$, PS's expected average rank is at most 6.4, whereas RP's is approximately 8.789.

The quota threshold. [Theorem 1](#) shows that PS's almost-constant rank guarantee holds uniformly over the number of copies, both in expectation and with high probability. Replication can nevertheless change its comparison with RP, because RP becomes better as the number of copies grows.

THEOREM 2. *Consider any sequence with $m \rightarrow \infty$ and integer quotas $q = q(m) \geq 1$.*

(1) *If $q = o(\log m)$, then*

$$\mathbb{E} \text{Rank}(\text{RP}) \sim \frac{\log m}{q}, \quad (12)$$

and

$$\mathbb{E} \text{Rank}(\text{RP}) - \mathbb{E} \text{Rank}(\text{PS}) \longrightarrow \infty. \quad (13)$$

(2) *If $q/\log m \rightarrow c \in (0, \infty)$, let $r_c > 1$ be the unique solution of*

$$r_c - 1 - \log r_c = \frac{1}{c}. \quad (14)$$

⁶For $911 \leq m \leq 3.8 \times 10^6$, $\log^* m = 3$, so PS's expected rank is at most 6.4, whereas RP's expected rank is increasing in m and exceeds 6.4 at $m = 911$. For larger m , use $\log^* m + 3.4 \leq 5.4 + \log \log m < \log m - 1 \leq \mathbb{E} \text{Rank}(\text{RP})$.

Then

$$\mathbb{E} \text{Rank}(\text{RP}) \longrightarrow r_c, \quad \limsup_{m \rightarrow \infty} \mathbb{E} \text{Rank}(\text{PS}) \leq 1 + \sqrt{\frac{2}{c}} < r_c. \quad (15)$$

In particular, the lower limit of the expected-rank gap is at least $r_c - 1 - \sqrt{2/c} > 0$.

(3) If $q/\log m \rightarrow \infty$, then

$$\mathbb{E} \text{Rank}(\text{PS}) \rightarrow 1, \quad \mathbb{E} \text{Rank}(\text{RP}) \rightarrow 1. \quad (16)$$

Consequently, along any such sequence,

$$|\mathbb{E} \text{Rank}(\text{RP}) - \mathbb{E} \text{Rank}(\text{PS})| \longrightarrow 0 \iff \frac{q}{\log m} \longrightarrow \infty. \quad (17)$$

The proof is in Appendix B.2. The proof also gives the stronger relative conclusion

$$\frac{\mathbb{E} \text{Rank}(\text{PS})}{\mathbb{E} \text{Rank}(\text{RP})} \longrightarrow 0 \quad \text{whenever } q = o(\log m). \quad (18)$$

Quotas of order $\log m$ make RP's expected rank bounded, whereas a vanishing gap from PS requires quotas to grow faster than $\log m$.

Figure 1 compares the two sequences at the same total market sizes, starting from 100 types with one copy each. Panel (a) holds $q = 1$ fixed and increases m , whereas panel (b) holds $m = 100$ fixed and increases q . The first sequence illustrates the widening gap when growth adds distinct alternatives, whereas the second shows RP catching up as both mechanisms improve towards rank one.

Why RP catches up. With m fixed, the number of agents ranking a given type first has mean q and variance $q(1 - 1/m)$. First-choice demand therefore fluctuates on the scale $\sqrt{q}$, while capacity grows on the scale q . Formally, (44) and (96) imply

$$\lim_{q \rightarrow \infty} \mathbb{E} \text{Rank}(\text{PS}) = \lim_{q \rightarrow \infty} \mathbb{E} \text{Rank}(\text{RP}) = 1 \quad \text{for fixed } m.$$

This fixed- m limit uses our uniform preferences and equal quotas. It illustrates the Che–Kojima equivalence in this model, whereas their theorem applies more generally, including markets in which some types remain disproportionately popular.

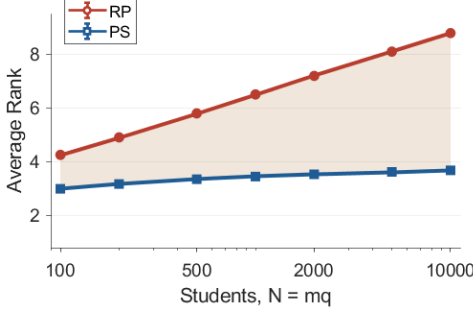
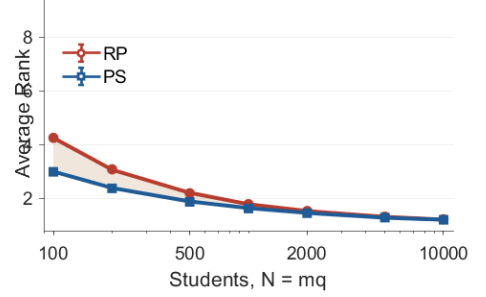

 (a) $q = 1, n = m$.

 (b) $m = 100, q = n/100$.

Fig. 1. PS and RP under two ways of growing the market. Both panels use $n = mq \in \{100, 200, 500, 1000, 2000, 5000, 10000\}$ agents and goods and i.i.d. uniform rankings over types. 2,000 markets per n . The mechanisms use independent samples, and error bars are pointwise 95% Monte Carlo confidence intervals.

4 Proof of Theorem 1

The upper and lower bounds use the same stream of independent draws to track consumption. For the upper bound, few types can remain only after many draws, by which point most surviving types have many consumers. Their limited supply bounds each phase's contribution to average rank. For the lower bound, types not yet drawn retain their entire supply; consuming that stock forces nearly one unit of expected rank at each successive scale. In both arguments the scales are related by repeated exponentiation, yielding an iterated-logarithmic number of phases. We first give these arguments, then sharpen the expectation bound to obtain the explicit constant and the quota comparison.

4.1 Generating preferences and counting draws

The case $m = 1$ is immediate, so assume $m \geq 2$. Write $r_g(t)$ for type g 's remaining supply and define

$$\mathcal{A}(t) := \{g : r_g(t) > 0\}, \quad A(t) := |\mathcal{A}(t)|. \quad (19)$$

Let $c_g(t)$ be the number of agents consuming type g . Between choices, supply falls at the rate $r'_g(t) = -c_g(t)$. At an exhaustion time, remove all types exhausted

simultaneously and process the displaced agents' choices in a fixed order, without advancing time. Quantities at that time refer to the state after these choices.

LEMMA 2. *The preference orders can be revealed sequentially so that, whenever an agent needs to choose, her choice is uniform among the surviving types conditional on the history revealed so far.*

PROOF. Give each agent an independent stream of uniform draws from G , and rank types by their first appearances in her stream. Symmetry gives independent uniform preference orders. Whenever an agent needs a choice, read her stream until a surviving type appears. Every type she has previously encountered is then exhausted: earlier rejected types were already unavailable, and every previously accepted type was consumed until exhaustion. The first surviving label is therefore her favourite available type.

Interleave the streams in the order PS requests choices. Which agent draws next is determined before her next label is seen, so the interleaved labels remain independent and uniform. This is Knuth's principle of deferred decisions in this setting [Knuth, 1976]. $\square$

For example, the stream $a, a, b, c, b, d, \dots$ generates the ranking $a \succ b \succ c \succ d$. After consuming a until exhaustion, the agent skips the repeated a and reaches b . Her current rank is two, although she has read three labels. Generally, her current rank is the number of distinct types encountered and is at most the number of labels read.

Write $\xi_1, \xi_2, \dots$ for the interleaved labels and let $K(t)$ be the total number read by time $t < 1$, after all agents have chosen. For $S \subseteq G$, define its count in the first k draws by

$$L_S(k) := \sum_{\ell=1}^k \mathbf{1}\{\xi_\ell \in S\}. \quad (20)$$

Every appearance of a type that still survives brought one of its current consumers. Hence

$$c_g(t) = L_{\{g\}}(K(t)) \quad (g \in \mathcal{A}(t)), \quad L_{\mathcal{A}(t)}(K(t)) = n. \quad (21)$$

If a prefix ends during simultaneous reassignments, the surviving types account for at most n labels; equality holds once all agents have chosen.

For a nonempty set S , write $\rho_i(S) := \min_{g \in S} \text{rank}_i(g)$. Assignment shares are consumption times, so

$$\text{Rank}(\text{PS}) = \int_0^1 \frac{1}{n} \sum_{i=1}^n \rho_i(\mathcal{A}(t)) dt \leq \frac{1}{n} \int_0^1 K(t) dt. \quad (22)$$

The inequality holds for every realization. It reduces the upper bound on rank to a bound on draws.

For any fixed set S of size s , $L_S(k)$ is binomial with mean ks/m . The standard lower-tail Chernoff bound gives

$$\mathbb{P}(L_S(\lceil 4nm/s \rceil) \leq n) \leq e^{-n(3-\log 4)}.$$

Since $3 - \log 4 > \log 2$ and $m \leq n$, a union bound over at most 2^m sets shows that, except with exponentially small probability,

$$L_S(\lceil 4nm/|S| \rceil) > n \quad \text{for every nonempty } S. \quad (23)$$

On this event, (21) implies $K(t) < 4nm/A(t)$, and therefore

$$\text{Rank}(\text{PS}) \leq 4 \int_0^1 \frac{m}{A(t)} dt. \quad (24)$$

The ratio $m/A(t)$ measures the scarcity of available choices. It remains to show that PS spends little time with few types left.

4.2 The upper bound: crowded types exhaust quickly

The same draw counts control the number of consumers. Once few types remain, those types must account for all n agents, requiring many draws. Most surviving types have then appeared many times and therefore have many consumers.

PROPOSITION 1. *There are constants $\eta_0, \alpha > 0$ and $\Lambda_0 < \infty$, independent of m and q , such that the following holds with probability at least $1 - O(n^{-3})$. Define*

$$b(s) := \left\lceil m \exp\left(-\alpha \frac{m}{s}\right) \right\rceil. \quad (25)$$

At every time with $A(t) = s \leq m/\Lambda_0$, fewer than $b(s)$ surviving types have fewer than $\eta_0 n/(4s)$ consumers. Consequently, every $T \subseteq \mathcal{A}(t)$ with $|T| \geq 2b(s)$ satisfies

$$\sum_{g \in T} c_g(t) \geq \frac{\eta_0}{8} \frac{n|T|}{s}. \quad (26)$$

PROOF. Choose a small constant $\eta_0 > 0$ and put $k_s := \lfloor \eta_0 nm/s \rfloor$. For a fixed set S of size s , $L_S(k_s)$ has mean at most $\eta_0 n$. Chernoff's bound gives

$$\mathbb{P}(L_S(k_s) \geq n) \leq \exp[-n(\log(1/\eta_0) - 1 + \eta_0)].$$

Choose η_0 so that the exponent exceeds $n \log 2$. A union bound over all sets then shows, except with exponentially small probability, that

$$A(t) = s \implies K(t) > k_s. \quad (27)$$

Indeed, fewer draws could not give the surviving set its n consumers. The binomial estimate has a simple interpretation: each fixed type has expected count of order $n/s = qm/s$ by this prefix, and a count below half its mean has probability exponentially small in qm/s .

Now fix a set T of $b = b(s)$ types. For $\Lambda_0 \geq 2/\eta_0$, its count in the first k_s draws has mean at least $\eta_0 bn/(2s)$. If all its types survive with fewer than $\eta_0 n/(4s)$ consumers each, that count must be less than half its mean. Another Chernoff bound gives probability at most

$$\exp\left(-\frac{\eta_0 bn}{16s}\right). \quad (28)$$

Taking a union bound over the possible T and s proves the assertion for sufficiently small α . Appendix A.1 gives the constants and probability calculation, together with the version used in the extensions. Finally, any group of at least $2b(s)$ surviving types has at least half its members above the consumer threshold, which gives (26). $\square$

Supply bounds each phase's contribution. Work on the event in Proposition 1. A phase beginning with ℓ surviving types ends once at most $2b(\ell)$ remain. The types surviving throughout this phase have limited supply, while the proposition bounds their consumption rate below.

PROPOSITION 2. *On the event in Proposition 1,*

$$\int_0^1 \frac{m}{A(t)} dt = O(\log^* m). \quad (29)$$

PROOF. Increase Λ_0 so that, for $\lambda \geq \Lambda_0$, $2\lambda e^{-\alpha\lambda} \leq 1/4$ and $\alpha\lambda/2 \geq \log(8/\alpha)$. These conditions ensure that $2b(\ell) < \ell$ whenever $3 \leq \ell \leq m/\Lambda_0$, and that the phase thresholds grow sufficiently quickly.

Before $t_0 := \inf\{t : A(t) \leq m/\Lambda_0\}$, the integral contributes at most Λ_0 . Starting at $t_j < 1$ with $\ell_j := A(t_j) > 2$, define

$$t_{j+1} := \inf\{t > t_j : A(t) \leq 2b(\ell_j)\}.$$

Let $T_j := \mathcal{A}(t_{j+1}-)$ be the types surviving until just before the phase ends. Throughout the phase, $T_j \subseteq \mathcal{A}(t)$ and $|T_j| > 2b(\ell_j) \geq 2b(A(t))$. Their initial remaining supply is at most $q|T_j|$. Since consumption cannot exceed this supply, (26) gives

$$q|T_j| \geq \int_{t_j}^{t_{j+1}} \sum_{g \in T_j} c_g(t) dt \geq \frac{\eta_0 q |T_j|}{8} \int_{t_j}^{t_{j+1}} \frac{m}{A(t)} dt. \quad (30)$$

Thus each phase contributes at most $8/\eta_0$. The quota cancels because increasing supply also increases the number of consumers.

It remains to count the phases. Put $\lambda_j := m/\ell_j$. Unless the next phase is terminal, the rounded threshold gives

$$\lambda_{j+1} \geq \frac{1}{4} e^{\alpha\lambda_j}.$$

Indeed, if $m e^{-\alpha\lambda_j} \geq 1$, then $2b(\ell_j) \leq 4m e^{-\alpha\lambda_j}$; otherwise at most two types remain. With $u_j := \alpha\lambda_j/2$, our choice of Λ_0 gives

$$u_{j+1} \geq \frac{\alpha}{8} e^{2u_j} \geq e^{u_j}.$$

Repeated exponentiation reaches $\alpha\lambda_j > \log m$ in at most $\log^* m + O(1)$ steps. Once at most two types remain, their supply is at most $2q$, so the remaining time is at most $2/m$ and the terminal integral contributes at most two. Adding the initial segment and the phases proves the claim. $\square$

Combining (24) and Proposition 2, and adding the probabilities of failure of the counting estimates, proves (7). Since $\text{Rank}(\text{PS}) \leq m \leq n$, those exceptional events

contribute only $O(n^{-2}) + ne^{-cn}$ to expected rank. Hence $\mathbb{E} \text{Rank}(\text{PS}) = O(\log^* m)$ uniformly over q , for $m \geq 2$.

4.3 The lower bound: untouched supply must be consumed

Fix a positive integer q and $0 < \eta < 1/4$, and let $m \rightarrow \infty$. The upper argument showed that crowded types exhaust quickly. For the lower bound, consider types that have never been chosen: each still has its entire supply q . Before very little supply remains, most of this untouched stock must be consumed. We show that this forces nearly one unit of expected rank at each of an iterated-logarithmic number of scales.

Write σ_ℓ for the time of draw ℓ . Its value is determined before the label ξ_ℓ is seen. After PS finishes, append independent labels with draw times equal to one. Choose a fixed absolute constant $\gamma \in (0, 1)$ as specified in Appendix A.2, and define

$$b_m := \gamma \log m, \quad k_* := \lfloor mb_m \rfloor, \quad Z := \frac{1}{m} \int_0^1 \min\{K(t), k_*\} dt = \frac{1}{m} \sum_{\ell=1}^{k_*} (1 - \sigma_\ell).$$

Thus Z counts draws, weighted by the time remaining, and $Z \leq b_m$. Unlike in the upper bound, repeated draws cannot be counted as rank increases. Appendix A.2 makes the necessary comparison using the exact expected best rank in a fixed set of s types, $(m+1)/(s+1)$:

$$\mathbb{E} \text{Rank}(\text{PS}) \geq \frac{1 - \eta}{(1 + \eta)q} \mathbb{E} Z - O_{q,\eta}(1). \quad (31)$$

It remains to show that $\mathbb{E} Z$ is at least approximately $q \log^* m$.

Untouched supply. Choose a starting scale $x_0 > 1$ depending only on q and a multiplier $B > 1$ depending only on q and η , as specified in Appendix A.2. Define successive draw scales by

$$x_{j+1} := B e^{x_j}, \quad k_j := \lfloor m x_j \rfloor,$$

and let J be the largest index with $x_J \leq b_m$. For sufficiently large m , this index exists. For $j < J$, put

$$\mathcal{U}_j := G \setminus \{\xi_1, \dots, \xi_{k_j}\}, \quad Z_j := \frac{1}{m} \sum_{\ell=k_j+1}^{k_{j+1}} (1 - \sigma_\ell).$$

The set $\mathcal{U}_j$ consists of the types untouched after k_j draws. Its expected size is $m(1 - 1/m)^{k_j} \sim me^{-x_j}$. The variance estimate in Appendix A.2 shows that, with high probability, it contains at least $me^{-x_j}/2$ types.

On the intersection of this event with the event in (23), at most $4nm/k$ types survive after any prefix of k draws. Total supply remaining after prefix k_{j+1} is therefore at most

$$\frac{4nmq}{k_{j+1}} \leq \frac{8mq^2}{x_{j+1}} = \frac{8q}{B} qme^{-x_j} \leq \eta q|\mathcal{U}_j|.$$

The lower bound on untouched supply is proportional to e^{-x_j} , whereas the upper bound on remaining supply is proportional to $1/x_{j+1}$. Making the second a small fraction of the first explains our choice $x_{j+1} = Be^{x_j}$, with B sufficiently large relative to q/η . If PS has finished, remaining supply is zero. There are absolute constants $C, c > 0$ such that these estimates fail with probability at most $\varepsilon_m := Cm^{-c}$, uniformly in $j < J$. Thus, except with probability ε_m , at least $(1 - \eta)q|\mathcal{U}_j|$ of the initially untouched supply has been consumed by the next prefix. Each consumer of these types arrived at a draw in this interval, and a draw at time σ_ℓ contributes at most $1 - \sigma_\ell$ to consumption. Consequently,

$$\sum_{\ell=k_j+1}^{k_{j+1}} (1 - \sigma_\ell) \mathbf{1}\{\xi_\ell \in \mathcal{U}_j\} \geq (1 - \eta)q|\mathcal{U}_j| \quad (32)$$

on this event. A prefix ending during reassignments causes no difficulty because no time passes between those choices.

Converting consumption into rank. Every future draw hits $\mathcal{U}_j$ with conditional probability $|\mathcal{U}_j|/m$. Since the remaining time $1 - \sigma_\ell$ is known before that draw,

conditional expectation gives

$$\mathbb{E} \left[\frac{\mathbf{1}\{|\mathcal{U}_j| > 0\}}{|\mathcal{U}_j|} \sum_{\ell=k_j+1}^{k_{j+1}} (1 - \sigma_\ell) \mathbf{1}\{\xi_\ell \in \mathcal{U}_j\} \right] = \mathbb{E}[Z_j \mathbf{1}\{|\mathcal{U}_j| > 0\}] \leq \mathbb{E}Z_j,$$

with the fraction defined to be zero when $\mathcal{U}_j$ is empty. This identity holds before imposing the high-probability event. Applying (32) inside the expectation yields $\mathbb{E}Z_j \geq (1 - \eta)q(1 - \varepsilon_m)$. The draw intervals are disjoint and $k_j \leq k_*$, so

$$\frac{1}{q} \mathbb{E}Z \geq (1 - \eta)J(1 - \varepsilon_m). \quad (33)$$

Counting the scales. The recurrence $x_{j+1} = Be^{x_j}$ grows by repeated exponentiation. Comparing its iterates with exponential towers, as detailed in Appendix A.2, gives

$$J = \log^* m + O_{q,\eta}(1).$$

Combining (31) and (33), and noting that $J\varepsilon_m \rightarrow 0$, gives

$$\mathbb{E} \text{Rank}(\text{PS}) \geq \frac{(1 - \eta)^2}{1 + \eta} \log^* m - O_{q,\eta}(1).$$

First let $m \rightarrow \infty$ and then let $\eta \downarrow 0$. This proves the lower limiting ratio of one in (8); the explicit upper bound proved next gives the reverse inequality.

4.4 Explicit expected-rank bounds

A rank increase at time t affects only the fraction $1 - t$ of an agent's allocation still to be consumed. Accounting for this remaining fraction sharpens the expectation bound and gives the quota comparison.

List all accepted choices in chronological order, including the n initial choices at time zero. Choice h occurs at time τ_h , when a_h object types remain available. Each agent selects any object type at most once, so there are at most nm choices. Define

$$f_q(t) := t + \frac{1}{q} \mathbb{E} \sum_{\substack{h > n \\ \tau_h \leq t}} \frac{t - \tau_h}{a_h}, \quad 0 \leq t \leq 1. \quad (34)$$

At a choice with a_h types available, $1/a_h$ is the conditional probability of selecting any particular available type. Its contribution is weighted by the time $t - \tau_h$ during

which that choice can contribute to consumption before t . The initial choices contribute $nt/m = qt$ before division by capacity. Thus $f_q(t)$ is a deterministic measure of accumulated choices, weighted by their consumption time and divided by capacity.

LEMMA 3. *The expected average rank under PS satisfies*

$$\mathbb{E} \text{Rank(PS)} \leq 1 + \left(1 - \frac{1}{m}\right) (f_q(1) - 1) \leq f_q(1). \quad (35)$$

PROOF. For this calculation, reveal each agent's preference list only up to her current choice. Suppose the exhausted type has rank r and a types remain. All a surviving types lie in the agent's unrevealed suffix of length $m - r$. Conditional on the revealed history, their positions form a uniform a -element subset of that suffix. The expected position of the first is $(m - r + 1)/(a + 1)$: the selected positions divide the other $m - r - a$ positions into $a + 1$ exchangeable gaps. Therefore,

$$\mathbb{E}[\text{rank increase} \mid \text{revealed history}] = \frac{m - r + 1}{a + 1} \leq \frac{m}{a + 1}. \quad (36)$$

The initial consumed rank is one. A rank increase at time τ contributes that increase times $1 - \tau$ to the agent's average rank. Summing (36) over post-initial choices, averaging over n agents, and using $a \leq m - 1$ for all post-initial choices gives

$$\mathbb{E} \text{Rank(PS)} - 1 \leq \frac{1}{q} \mathbb{E} \sum_{h>n} \frac{1 - \tau_h}{a_h + 1} \leq \frac{1 - 1/m}{q} \mathbb{E} \sum_{h>n} \frac{1 - \tau_h}{a_h} = \left(1 - \frac{1}{m}\right) (f_q(1) - 1). \quad (37)$$

□

Average congestion. The quantity $f_q(1)$ also measures the number of consumers sharing an agent's chosen type:

$$f_q(1) = \frac{1}{nq} \mathbb{E} \int_0^1 \sum_{g \in G} c_g(t)^2 dt. \quad (38)$$

Indeed, $\sum_g c_g(t)^2/n$ averages this number across agents, including each agent herself. Thus $f_q(1)$ is average congestion over the eating process, divided by capacity.

Appendix A.3 verifies this identity.

The explicit constant. Appendix A.3 bounds increments of f_q directly. The argument compares the consumption implied by accumulated choices with the capacity of surviving types. A conditional exponential estimate controls the difference between choices and their conditional probabilities; concavity then turns it into a bound for each time interval.

For $m \geq 3$, each refined phase contributes at most one unit plus a remainder. The remainders have a bounded sum, and at most $\log^* m - 1$ phases are needed after the initial segment, including the final one. Appendix A.3.2 constructs these phases and bounds the initial segment and the remainders, giving

$$\mathbb{E} \text{Rank}(\text{PS}) \leq f_q(1) < \log^* m + 3.4. \quad (39)$$

For $m = 1, 2$, average rank is at most m . This proves the explicit expectation bound in Theorem 1.

Remaining supply and the quota bound. Define $\mathcal{W}(t) := \sum_g r_g(t) c_g(t)$. Since $r'_g(t) = -c_g(t)$ almost everywhere and each type's supply falls from q to zero,

$$\int_0^1 \mathcal{W}(t) dt = -\frac{1}{2} \int_0^1 \frac{d}{dt} \sum_g r_g(t)^2 dt = \frac{mq^2}{2}. \quad (40)$$

Every type contributes $q^2/2$ to this integral, regardless of when its capacity is consumed.

The same exponential estimate, together with (40), yields two inequalities, proved in Appendix A.4: for all $\theta, \eta > 0$,

$$f_q(1) \leq 1 + \frac{\log m}{q\theta} + \theta \int_0^1 f_q(t) dt, \quad (41)$$

$$\int_0^1 f_q(t) dt \leq \frac{\eta + 1/(2q)}{2(1 - e^{-\eta})}. \quad (42)$$

The first inequality controls the final value by the accumulated values earlier in the process; its logarithmic term allows for the m possible identities of the last surviving type. The second uses the supply identity to bound those earlier values

independently of the number of types. Combining them gives

$$f_q(1) \leq 1 + \frac{\log m}{q\theta} + \theta \frac{\eta/2 + 1/(4q)}{1 - e^{-\eta}}. \quad (43)$$

Setting $\eta = q^{-1/2}$ and minimizing over $\theta > 0$ yields, by [Lemma 3](#),

$$\mathbb{E} \text{Rank}(\text{PS}) \leq 1 + \left(1 - \frac{1}{m}\right) \sqrt{\frac{2 \log m}{q}} \left(1 + \frac{0.55}{\sqrt{q}}\right). \quad (44)$$

The parameter calculation is in [Appendix A.4](#). This is the PS estimate used in [Theorem 2](#): at large quotas its excess rank is bounded by a term of order $\sqrt{\log m/q}$. These expectation arguments require only $q \geq 1$, so they also apply when the common capacity is real and $n = mq$ is an integer.

5 Extensions

5.1 Unequal quotas

Let each type $g \in G$ have quota $q_g \geq 1$, and let $n = \sum_{g \in G} q_g$ agents have independent uniform rankings over the m types. Write $\bar{q} = n/m$ for the average quota and $q_{\max} = \max_g q_g$ for the largest quota.

PROPOSITION 3. *Fix $K \geq 1$ and suppose $q_{\max} \leq K\bar{q}$. There are constants $C_K, c_K > 0$, depending only on K , such that*

$$\mathbb{P}(\text{Rank}(\text{PS}) > C_K(1 + \log^* m)) \leq C_K n^{-3} + e^{-c_K n}. \quad (45)$$

Consequently, $\mathbb{E} \text{Rank}(\text{PS}) = O_K(\log^* m)$ as $m \rightarrow \infty$.

Expected initial demand per type equals the average quota, while remaining supply is bounded using the largest quota. Their ratio changes the phase bound only by a factor K . The proof appears in [Appendix C.1](#).

5.2 Unequal numbers of agents and goods

For the next two results, let there be n agents and m unit-capacity goods, with independent uniform preferences. Under PS, agents eat at unit speed until either every agent has consumed one unit or all goods have been exhausted. Under RP,

agents choose in a uniform random order until either all agents have chosen or all goods have been assigned. A feasible assignment is a non-negative $n \times m$ matrix satisfying

$$\sum_{g \in G} x_{ig} \leq 1 \quad \text{for every } i, \quad \sum_{i \in I} x_{ig} \leq 1 \quad \text{for every } g, \quad \sum_{i \in I} \sum_{g \in G} x_{ig} = \min\{n, m\}. \quad (46)$$

Thus every agent receives one unit when $m \geq n$, and all goods are allocated when $n \geq m$.

More goods than agents. When $m \geq n$, average rank retains its original definition in (4).

THEOREM 3. Fix $B < \infty$ and consider any sequence $n \rightarrow \infty$ with $n \leq m = m(n) \leq Bn$. Put

$$\Delta := m - n, \quad \Delta_+ := \Delta \vee 1. \quad (47)$$

Then

$$\mathbb{E} \text{Rank}(\text{PS}) = O_B \left(\log^* \left(1 + \frac{n}{\Delta_+} \right) \right), \quad (48)$$

whereas

$$\mathbb{E} \text{Rank}(\text{RP}) = \frac{m+1}{n} (H_{m+1} - H_{\Delta+1}). \quad (49)$$

Consequently,

$$\frac{\mathbb{E} \text{Rank}(\text{PS})}{\mathbb{E} \text{Rank}(\text{RP})} \rightarrow 0 \iff m - n = o(n). \quad (50)$$

If $(m - n)/n \rightarrow c > 0$, then

$$\mathbb{E} \text{Rank}(\text{RP}) \rightarrow (1 + c) \log \frac{1 + c}{c}. \quad (51)$$

Here B bounds supply relative to demand, and the PS bound's constant may depend on B . Thus a growing surplus need not wash away the divergence between PS and RP. A proportional surplus does make RP's expected rank bounded. The proof is in Appendix C.2.1.

More agents than goods. When $n \geq m$, both mechanisms leave total unassigned mass $n - m$. To compare the goods allocated, define average rank per allocated unit

by

$$\text{Rank}^{\text{alloc}}(M; \succ) := \frac{1}{m} \sum_{i=1}^n \sum_{g=1}^m \text{rank}_i(g) M_{ig}(\succ). \quad (52)$$

Write $\text{Rank}^{\text{alloc}}(M)$ for the corresponding random variable when preferences are drawn.

THEOREM 4. *For every $n \geq m \geq 1$,*

$$\mathbb{E} \text{Rank}^{\text{alloc}}(\text{PS}) \leq \min \left\{ \log^* m + 3.4, 1 + \sqrt{\frac{2m \log m}{n}} + \frac{m \log m}{n} \right\}, \quad (53)$$

whereas

$$\mathbb{E} \text{Rank}^{\text{alloc}}(\text{RP}) = \frac{m+1}{m} (H_{m+1} - 1) = \log m + O(1). \quad (54)$$

Consequently, along every sequence $n \geq m \rightarrow \infty$,

$$\frac{\mathbb{E} \text{Rank}^{\text{alloc}}(\text{PS})}{\mathbb{E} \text{Rank}^{\text{alloc}}(\text{RP})} \rightarrow 0. \quad (55)$$

Moreover, for every fixed $C > 0$, the PS bound is $O_C(1)$ whenever $n/m \geq C \log m$; in particular, for every fixed $\varepsilon > 0$, it is $O_\varepsilon(1)$ whenever $n \geq m^{1+\varepsilon}$.

The proof is a time-and-capacity rescaling of the balanced market and appears in Appendix C.2.2.

If instead of measuring the rank of only real objects, we could artificially assign rank $m+1$ to being unassigned. In that case, average rank becomes

$$\text{Rank}^{\text{out}}(M) = \frac{m}{n} \text{Rank}^{\text{alloc}}(M) + \left(1 - \frac{m}{n}\right) (m+1). \quad (56)$$

As $m \rightarrow \infty$, if m/n stays bounded away from one, the ratio of these per-agent expected ranks tends to one. The common second term is at least a positive constant times m , whereas the expected assigned-good component is $O(\log m)$ under RP and $O(\log^* m)$ under PS. This is why we think average rank per allocated unit is the more informative measure of assignment quality.

5.3 Correlated preferences

Return to the one-copy case $q = 1$, with $n = m$ agents and m unit-capacity goods.

DEFINITION 1. Let $w = (w_g)_{g \in G}$ be a possibly random vector of positive popularity weights, common to all agents, satisfying almost surely

$$\frac{\max_g w_g}{\min_g w_g} \leq \kappa \quad (57)$$

for a fixed $\kappa < \infty$. Conditional on w , agents draw independent Plackett–Luce rankings: each successive good in a ranking is chosen from the remaining goods with probability proportional to its weight.

Random common weights can therefore correlate preferences across agents. Expectations below include both the common weights and the rankings drawn conditional on them.

THEOREM 5. As $m \rightarrow \infty$, under *Definition 1*,

$$\mathbb{E} \text{Rank}(\text{PS}) = O_\kappa(\log^* m), \quad \mathbb{E} \text{Rank}(\text{RP}) = \Theta_\kappa(\log m). \quad (58)$$

Consequently,

$$\frac{\mathbb{E} \text{Rank}(\text{PS})}{\mathbb{E} \text{Rank}(\text{RP})} \longrightarrow 0. \quad (59)$$

The proof appears in Appendix C.3. This result requires the stated restriction on dependence: if all agents share the same uniformly drawn ranking, both mechanisms have average rank $(m + 1)/2$ by *Proposition 4*.

6 Conclusion

Random assignment involves a choice among incompatible desiderata. Random Priority is strategy-proof but not ordinally efficient, whereas Probabilistic Serial is ordinally efficient but not strategy-proof (and these properties are incompatible if equal treatment of equals is also required). Whether giving up strategy-proofness is worth it has been answered by appeal to large-market equivalences. Our results suggest that the answer critically depends on the market. When objects are many and copies are few relative to the logarithm of the number of types, the ordinal efficiency of Probabilistic Serial translates into markedly better assignments. Only when copies per type vastly exceed that logarithm the choice of mechanism becomes irrelevant.

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AI disclosure

We used AI to improve our upper bound, optimize its constants, and develop the lower-bound argument. We have verified the proofs and stand by the results. We also used AI assistance to produce a Lean formalization accompanying the upper-bound analysis of Theorem 1, which is available at www.josueortega.com, together with the Matlab code for Figure 1.

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A Auxiliary PS arguments

A.1 Counting draws and consumers

The same counting argument covers unequal supplies and bounded popularity weights. Capacities determine when types disappear, but do not enter the estimates below. The balanced model is the special case $D = \kappa = 1$.

LEMMA 4. *Fix $D, \kappa \geq 1$. Suppose $m \leq Dn$, and generate preferences by first appearances in independent streams with label probabilities p_g satisfying*

$$\frac{1}{\kappa m} \leq p_g \leq \frac{\kappa}{m}.$$

Suppose all n agents start consuming at time zero and continue at unit speed until a common stopping time, each keeping her chosen type until it is exhausted. There are constants $C, c, \Lambda, \eta, \alpha > 0$, depending only on D and κ , such that, with probability at least $1 - O_{D,\kappa}(n^{-3}) - e^{-cn}$, the following hold simultaneously before that stopping time, after choices at an exhaustion have been completed:

$$\frac{K(t)}{n} \leq C \frac{m}{A(t)}. \quad (60)$$

If $A(t) = s \leq m/\Lambda$, define

$$b(s) := \left\lceil m \exp\left(-\alpha \frac{m}{s}\right) \right\rceil. \quad (61)$$

Fewer than $b(s)$ surviving types have fewer than $\eta n/(4\kappa s)$ consumers. Consequently, every $T \subseteq \mathcal{A}(t)$ with $|T| \geq 2b(s)$ satisfies

$$\sum_{g \in T} c_g(t) \geq \frac{\eta}{8\kappa} \frac{n|T|}{s}. \quad (62)$$

The constants can be chosen so that $2b(s) < s$ whenever $3 \leq s \leq m/\Lambda$.

PROOF. It suffices to consider $m \geq 2$. Retain the prefix count $L_S(k)$ from [Section 4.1](#). For a fixed set S of size s , this count is binomial with success probability $p(S) := \sum_{g \in S} p_g$. The bounds on p_g give $s/(\kappa m) \leq p(S) \leq \kappa s/m$. We use the standard binomial estimates

$$\mathbb{P}(X \geq n) \leq e^{-n[\log(1/u) - 1 + u]} \quad \text{if } \mathbb{E}X \leq un < n, \quad \mathbb{P}(X \leq \mathbb{E}X/2) \leq e^{-\mathbb{E}X/8}. \quad (63)$$

For $\mathbb{E}X \geq an > n$, the corresponding lower-tail estimate is $\mathbb{P}(X \leq n) \leq e^{-n(a-1-\log a)}$.

The number of draws. Choose $C > \kappa$ so that $C/\kappa - 1 - \log(C/\kappa) > D \log 2 + 1/2$. At prefix length $\lceil Cnm/s \rceil$, the mean count of S is at least Cn/κ . A union bound over at most $2^m \leq 2^{Dn}$ sets therefore gives, except with probability $e^{-n/2}$,

$$L_S(\lceil Cnm/s \rceil) > n \quad \text{for every nonempty } S.$$

Whenever $S = \mathcal{A}(t)$, its count at $K(t)$ is exactly n . Hence $K(t) < \lceil Cnm/s \rceil$, which gives (60).

For the reverse comparison, choose $\eta > 0$ so small that $\kappa\eta < 1$ and $\log(1/(\kappa\eta)) - 1 + \kappa\eta > D \log 2 + 1/2$. Put

$$k_s := \left\lfloor \eta \frac{nm}{s} \right\rfloor.$$

The same union bound, now using the first estimate in (63), shows that $L_S(k_s) < n$ for every nonempty S , except with probability $e^{-n/2}$. Thus $K(t) > k_s$ whenever $A(t) = s$.

Types with few consumers. Write $\gamma := n/m \geq 1/D$, $\lambda := m/s$, and set

$$\beta := \frac{\eta}{16\kappa}, \quad \alpha := \frac{\beta}{16D}.$$

Choose $\Lambda \geq \max\{2/\eta, 1/\alpha\}$ sufficiently large that, for all $\lambda \geq \Lambda$,

$$1 + \alpha\lambda \leq \frac{\beta\lambda}{2D}, \quad 2\lambda e^{-\alpha\lambda} \leq \frac{1}{4}. \quad (64)$$

Fix $s \leq m/\Lambda$ and a set T of $b = b(s)$ types. Since $k_s \geq \eta nm/(2s)$, the binomial count $L_T(k_s)$ has mean at least $\eta nb/(2\kappa s)$. If all types in T survive and each has fewer than $\eta n/(4\kappa s)$ consumers, then, on the preceding prefix event, $L_T(k_s)$ is below half this mean. The probability of this intersection is at most

$$\exp\left(-\beta \frac{nb}{s}\right) = e^{-\beta\gamma b\lambda}.$$

There are at most $(em/b)^b$ candidate sets T . Since $b \geq me^{-\alpha\lambda}$, (64) bounds their total probability by $e^{-\beta\gamma b\lambda/2}$.

For $\lambda \geq \Lambda$, one has

$$b\lambda \geq \frac{\log m}{\alpha}. \quad (65)$$

Indeed, if $\alpha\lambda \geq \log m$, use $b \geq 1$; otherwise $m\lambda e^{-\alpha\lambda}$ is decreasing for $\alpha\lambda \geq 1$ and has value $(\log m)/\alpha$ at the right endpoint. Summing the failure bound over s therefore gives at most $m^{1-8D\gamma}$. This is at most n^{-3} : the ratio is $\gamma^3 m^{4-8D\gamma}$, which decreases for $\gamma \geq 1/D$ and $m \geq 2$, and at $\gamma = 1/D$ equals $D^{-3}m^{-4} \leq 1$.

Outside these exceptional events, fewer than $b(s)$ surviving types are underloaded. Any T of size at least $2b(s)$ thus has at least $|T|/2$ types with the stated number of consumers, giving (62). Finally, (64) gives $2b(s) \leq s/4 + 2 < s$ for $s \geq 3$. The estimates concern fixed sets and deterministic prefixes, so they hold at every stage of the adaptive eating process. $\square$

A.2 Auxiliary estimates for the lower bound

We give the parameter choices, rank comparison (31), untouched-type estimate, and scale count used in Section 4.3. Retain the fixed q , η , b_m , and Z from that subsection.

Parameter choices. Take

$$\gamma := \frac{1}{10}, \quad x_0 := 8q, \quad B := \frac{32q}{\eta}.$$

Thus $b_m = (\log m)/10$ and $x_{j+1} = (32q/\eta)e^{x_j}$, with the dependence stated in the main argument.

Rank and draw counts on small sets. For a fixed set S of s types, summing the tail $\mathbb{P}(\rho_i(S) > r) = \binom{m-r}{s} / \binom{m}{s}$ gives

$$\mu_s := \mathbb{E}\rho_i(S) = \frac{m+1}{s+1}, \quad X_i := \frac{\rho_i(S)}{\mu_s}, \quad \mathbb{E}X_i^2 = 1 + \frac{s(m-s)}{(s+2)(m+1)} \leq 2.$$

The X_i are independent with mean one. Using $e^{-u} \leq 1 - u + u^2/2$ and exponential Markov's inequality with parameter $\eta/2$ gives

$$\mathbb{P}\left(\frac{1}{n} \sum_i X_i < 1 - \eta\right) \leq e^{-\eta^2 n/4}.$$

For the label counts, the binomial estimate with parameter $-\log(1 + \eta)$ similarly gives

$$\mathbb{P}\left(L_S\left(\left\lceil \frac{(1 + \eta)nm}{s} \right\rceil\right) \leq n\right) \leq e^{-n(\eta - \log(1 + \eta))}.$$

Choose $\delta \in (0, 1/8)$ with

$$\delta \log(e/\delta) \leq \min\left\{\frac{\eta^2}{8}, \frac{\eta - \log(1 + \eta)}{2}\right\}.$$

There are at most $\exp[m\delta \log(e/\delta)]$ sets of size at most δm . A union bound, using $n = mq \geq m$, yields an event $\mathcal{E}_\eta$ with $\mathbb{P}(\mathcal{E}_\eta^c) \leq C_\eta e^{-c_\eta n}$ on which, simultaneously for all $1 \leq |S| \leq \delta m$,

$$L_S\left(\left\lceil \frac{(1 + \eta)nm}{|S|} \right\rceil\right) > n, \quad \frac{1}{n} \sum_i \rho_i(S) \geq (1 - \eta) \frac{m + 1}{|S| + 1}. \quad (66)$$

No lower cutoff on $|S|$ is needed.

Set $w(t) := \min\{K(t), k_*\}/m$. If $A(t) \leq \delta m$, the first estimate and (21) give $w(t) < (1 + \eta)n/A(t)$. The second therefore implies

$$\frac{1}{n} \sum_i \rho_i(\mathcal{A}(t)) \geq \frac{(1 - \eta)w(t)}{(1 + \eta)q + w(t)/m} \geq \frac{(1 - \eta)w(t)}{(1 + \eta)q + b_m/m}.$$

If $A(t) > \delta m$, apply the first estimate to a surviving subset of size $\lfloor \delta m \rfloor$. For sufficiently large m , this gives $w(t) \leq 2(1 + \eta)q/\delta$. Hence (22) yields, on $\mathcal{E}_\eta$,

$$\text{Rank}(\text{PS}) \geq \frac{1 - \eta}{(1 + \eta)q + b_m/m} Z - \frac{2}{\delta}.$$

Since $Z \leq b_m$, replacing the coefficient by $(1 - \eta)/[(1 + \eta)q]$ changes the lower bound by at most $O_{q,\eta}(b_m^2/m) = o(1)$. The exceptional event contributes at most $O_{q,\eta}(b_m e^{-c_\eta n}) = o(1)$ when taking expectations. This proves (31).

The number of untouched types. For $x_j \leq b_m$, let $\mu_j := \mathbb{E}|\mathcal{U}_j| = m(1 - 1/m)^{k_j}$. Uniformly over these indices, $\mu_j \geq 0.9me^{-x_j}$ for sufficiently large m . Two types are both untouched with probability $(1 - 2/m)^{k_j} \leq (1 - 1/m)^{2k_j}$, so their indicators have nonpositive covariance and $\text{Var}(|\mathcal{U}_j|) \leq \mu_j$. Chebyshev's inequality gives

$$\mathbb{P}\left(|\mathcal{U}_j| < \frac{1}{2}me^{-x_j}\right) \leq \frac{\mu_j}{(\mu_j - me^{-x_j}/2)^2} \leq \frac{6e^{x_j}}{m} \leq 6m^{-9/10}.$$

Supply and failure probability. On the event in (23) and the untouched-type event above, the total supply remaining after prefix k_{j+1} is at most

$$\frac{4nmq}{k_{j+1}} \leq \frac{8mq^2}{x_{j+1}} = \frac{\eta}{4} qme^{-x_j} \leq \eta q |\mathcal{U}_j|.$$

Here $k_{j+1} \geq mx_{j+1}/2$ for sufficiently large m . The two events fail with probability at most $C'e^{-c'n} + 6m^{-9/10}$, where $C', c' > 0$ are absolute. Since $n = mq \geq m$, this is at most $Cm^{-9/10}$ for a suitable absolute constant C . Thus one may take $c := 9/10$ in the definition of ε_m in the main argument.

Counting the scales. Put $\zeta := 1 + \log B$. The recurrence satisfies $e^{x_j} \leq x_{j+1}$ and $x_{j+1} + \zeta \leq e^{x_j + \zeta}$. Hence, writing $\exp^{\circ j}$ for j successive exponentiations,

$$\exp^{\circ j}(8q) \leq x_j \leq \exp^{\circ j}(8q + \zeta).$$

Together with $x_j \leq b_m < x_{j+1}$, this gives $\log^* b_m - \log^*(8q + \zeta) - 1 \leq J \leq \log^* m$. Since $\log \log m \leq b_m \leq \log m$ for sufficiently large m , $\log^* b_m = \log^* m + O(1)$. For fixed q and η , therefore, $J = \log^* m + O_{q,\eta}(1)$, and the polynomial failure bound ensures $J\varepsilon_m \rightarrow 0$.

A.3 The explicit expectation bound

We use the chronological choices τ_h, a_h and the function f_q defined in Section 4.4. Throughout this subsection and the quota argument below, q may be any real number at least one, provided $n = mq$ is an integer.

Verifying the congestion identity. Recall $\mathcal{W}(t) := \sum_g r_g(t) c_g(t)$ from the remaining-supply argument in Section 4.4. Immediately after the initial choices, $\mathcal{W}(0) = nq$. Between choices, $\mathcal{W}'(t) = -\sum_g c_g(t)^2$. Removing exhausted types leaves $\mathcal{W}$ unchanged because their remaining supply is zero. Each subsequent choice adds the remaining supply of the selected type, whose conditional expectation is $n(1 - \tau_h)/a_h$. Since $\mathcal{W}(1) = 0$, summing these changes gives

$$\mathbb{E} \int_0^1 \sum_g c_g(t)^2 dt = nq + n\mathbb{E} \sum_{h>n} \frac{1 - \tau_h}{a_h} = nq f_q(1). \quad (67)$$

A.3.1 A conditional exponential bound. Fix a deterministic horizon $t \leq 1$. For each type g , record choices up to time t and stop recording when it is exhausted. Let X_h^g indicate a recorded entry into g , and let p_h^g be its conditional probability given the preceding history. This probability is $1/a_h$ while g is available and the choice occurs by t , and zero otherwise. For any non-negative weights z_h determined before choice h and any $\theta > 0$, successive conditioning gives

$$M_g := \exp \left\{ -\theta \sum_h z_h X_h^g + \sum_h p_h^g (1 - e^{-\theta z_h}) \right\}, \quad \mathbb{E} M_g \leq 1. \quad (68)$$

Indeed, put $d_h := 1 - e^{-\theta z_h}$. The conditional expectation of the new factor is $e^{p_h^g d_h} (1 - p_h^g d_h) \leq 1$. Conditioning successively proves the bound; there are at most nm choices, and the sequence can be padded with zero terms.

Take $z_h = t - \tau_h$ for choices made by t , and zero otherwise. For every type still available at t , the positive term in the exponent is

$$F_{\theta,t} := q(1 - e^{-\theta t}) + \sum_{\substack{h > n \\ \tau_h \leq t}} \frac{1 - e^{-\theta(t - \tau_h)}}{a_h}. \quad (69)$$

Its weighted entry total is its consumed capacity, $q - r_g(t)$. Put $u := 1 - t > 0$ and $a := A(t)/m$. Since $\sum_g r_g(t) = mqu$, Jensen's inequality over surviving types gives

$$\sum_{g \in G} M_g \geq ma \exp \left\{ F_{\theta,t} - q\theta \left(1 - \frac{u}{a} \right) \right\}. \quad (70)$$

The sum on the left includes all types, with recording stopped at exhaustion. Taking logarithms, using $a \leq 1$ and $q \geq 1$, and setting $z := \theta u/a$, we obtain

$$\begin{aligned} \frac{F_{\theta,t}}{q} &\leq \theta \left(1 - \frac{u}{a} \right) - \frac{\log a}{q} + \frac{1}{q} \log \left(\frac{1}{m} \sum_g M_g \right) \\ &\leq \theta - \log(\theta u) + \log z - z + \frac{1}{q} \log \left(\frac{1}{m} \sum_g M_g \right). \end{aligned} \quad (71)$$

Now $\log z - z \leq -1$, and Jensen's inequality and (68) imply $\mathbb{E} \log(\sum_g M_g/m) \leq 0$. Consequently,

$$\frac{1}{q} \mathbb{E} F_{\theta,t} \leq \theta - 1 - \log(\theta(1 - t)), \quad 0 \leq t < 1, \quad \theta > 0. \quad (72)$$

This avoids conditioning on the random surviving set.

At time one, choose a type exhausted exactly then. Its weighted entry total is q and its positive exponent term is $F_{\theta,1}$, so $e^{F_{\theta,1}-q\theta} \leq \sum_g M_g$. Therefore

$$\frac{1}{q} \mathbb{E} F_{\theta,1} \leq \theta + \frac{\log m}{q}. \quad (73)$$

The sum over types allows for the random identity of the last surviving type.

A.3.2 A direct bound for each phase. Fix $0 \leq s < t \leq 1$ and put $\Delta := t - s$. A choice made by s contributes Δ/a_h to the increase in the weighted sum defining f_q ; a choice made between s and t contributes $(t - \tau_h)/a_h$. Concavity of $1 - e^{-\theta z}$ therefore gives, pathwise,

$$F_{\theta,t} \geq \frac{1 - e^{-\theta\Delta}}{\Delta} \left\{ \Delta \sum_{\tau_h \leq s} \frac{1}{a_h} + \sum_{s < \tau_h \leq t} \frac{t - \tau_h}{a_h} \right\}. \quad (74)$$

Here both sums include initial choices when applicable. The expected expression in braces is $q(f_q(t) - f_q(s))$. Applying (72) yields

$$f_q(t) - f_q(s) \leq \frac{(t - s) [\theta - 1 - \log(\theta(1 - t))]}{1 - e^{-\theta(t-s)}}, \quad t < 1. \quad (75)$$

For the final interval, (73) and $q \geq 1$ give

$$f_q(1) - f_q(1 - u) \leq \frac{\theta u + u \log m}{1 - e^{-\theta u}}, \quad 0 < u \leq 1. \quad (76)$$

The initial estimate follows by setting $s = 0$ and $\theta = 1 + t$ in (75):

$$f_q(t) \leq \frac{t [t - \log(1 - t^2)]}{1 - e^{-t-t^2}}, \quad 0 < t < 1. \quad (77)$$

These three inequalities are all that the following phase construction needs.

LEMMA 5. *For $m \geq 3$ and $q \geq 1$, put $z_j := (5/6)2^{-j}$ for $j \geq 0$. There are an integer $J \geq 0$ and times $0 < s_0 < \dots < s_J < 1$ such that $J + 1 \leq \log^* m - 1$,*

$$\begin{aligned} f_q(s_0) &< 2.108, \\ f_q(s_{j+1}) - f_q(s_j) &\leq e^{z_j}, \quad 0 \leq j < J, \\ f_q(1) - f_q(s_J) &\leq e^{z_J}. \end{aligned} \quad (78)$$

Moreover, $\sum_{j=0}^{\infty} (e^{z_j} - 1) < 2.291$.

PROOF. *Bounding each phase's contribution.* Define

$$z_j := \frac{5}{6} 2^{-j}, \quad c_j := e^{z_j} - 1 - z_j, \quad \lambda_0 := \frac{e}{c_0}, \quad \lambda_{j+1} := e z_j \lambda_j \exp(c_j \lambda_j). \quad (79)$$

Write $x_j := c_j \lambda_j$, $u_j := 1/\lambda_j$, and $s_j := 1 - u_j$. We first check that

$$x_j \geq e 2^j, \quad x_{j+1} \geq e^{x_j}. \quad (80)$$

Indeed, if $c(z) := e^z - 1 - z$, then

$$c(z) = z^2 \int_0^1 (1-r) e^{zr} dr, \quad \frac{c(z/2)}{c(z)} \geq \frac{1}{4} e^{-z/2}. \quad (81)$$

Thus, assuming $x_j \geq e 2^j$,

$$\begin{aligned} x_{j+1} &= \frac{c_{j+1}}{c_j} e z_j x_j e^{x_j} \\ &\geq \frac{1}{4} e^{-5/12} e \frac{5}{6} 2^{-j} (e 2^j) e^{x_j} = \frac{5}{24} e^{19/12} e^{x_j} \geq e^{x_j}. \end{aligned} \quad (82)$$

Since $e^x \geq 2x$ for $x > 0$ and $x_0 = e$, this proves (80) by induction. It also implies that λ_j increases and $0 < s_j < s_{j+1} < 1$.

For phase j , choose $\theta_j := z_j \lambda_j$. The definition of λ_{j+1} gives $\theta_j u_{j+1} = e^{-1-x_j}$, and hence

$$\theta_j - 1 - \log(\theta_j u_{j+1}) = \theta_j + x_j = \frac{z_j + c_j}{u_j}. \quad (83)$$

The function $d/(1 - e^{-\theta d})$ increases in $d > 0$. Since $s_{j+1} - s_j \leq u_j$, (75) therefore yields

$$f_q(s_{j+1}) - f_q(s_j) \leq \frac{z_j + c_j}{1 - e^{-z_j}} = e^{z_j}. \quad (84)$$

Stopping the phases. For $m \geq 3$, let J be the first index with $x_J \geq \log m$. Apply (76) with $u = u_J$ and $\theta = z_J/u_J$. Since $u_J \log m \leq c_J$,

$$f_q(1) - f_q(s_J) \leq \frac{z_J + c_J}{1 - e^{-z_J}} = e^{z_J}. \quad (85)$$

For the number of phases, let $E_0 := 1$ and $E_{k+1} := e^{E_k}$. Equation (80) implies $x_j \geq E_{j+1}$. If $k := \log^* m$, then $k \geq 2$ and $\log m \leq E_{k-1}$. Consequently $J \leq k - 2$, or $J + 1 \leq \log^* m - 1$. This gives the required number of phases.

All constants are explicit. Put $z := 5/6$. For the series,

$$\sum_{j=0}^{\infty} (e^{z2^{-j}} - 1) = \sum_{k=1}^{\infty} \frac{z^k}{k!(1 - 2^{-k})} \leq 2z + \frac{4}{3}(e^z - 1 - z) < 2.291. \quad (86)$$

For the initial value,

$$s_0 = 1 - \frac{e^{5/6} - 1 - 5/6}{e}, \quad \frac{s_0 [s_0 - \log(1 - s_0^2)]}{1 - e^{-s_0 - s_0^2}} < 2.108. \quad (87)$$

The last bound follows, for example, from $.82796 < s_0 < .82797$: evaluate the numerator in (77) at .82797 and its denominator at .82796. This proves the bounds on the initial segment and the sum of remainders. $\square$

Combining Lemmas 3 and 5, for $m \geq 3$ we obtain

$$\mathbb{E} \text{Rank}(\text{PS}) \leq f_q(1) < 2.108 + (\log^* m - 1) + 2.291 = \log^* m + 3.399 < \log^* m + 3.4.$$

A.4 The PS bound with quotas

We prove (41) and (42), and then choose the parameters to obtain (44). The case $m = 1$ is immediate, so assume $m \geq 2$.

PROOF. *Step 1: weight choices by the allocation still outstanding.* Integrating the definition of f_q term by term, including the initial choices, gives

$$\mathbb{E} \sum_h \frac{(1 - \tau_h)^2}{a_h} = 2q \int_0^1 f_q(t) dt. \quad (88)$$

The inequality $1 - e^{-x} \geq x - x^2/2$ and (69) therefore imply

$$\mathbb{E} F_{\theta,1} \geq \theta q f_q(1) - \theta^2 q \int_0^1 f_q(t) dt. \quad (89)$$

Combining this with the endpoint estimate (73) proves (41).

Step 2: use remaining supply to bound the integral. Fix $t < 1$ and $\eta > 0$, and put $u := 1 - t$ and $b := 1 - e^{-\eta}$. Apply (68) to choices made by t with unit weights and parameter η , denoting the resulting quantities by $\tilde{M}_g(t)$. For a surviving type,

$$\tilde{M}_g(t) = \exp \left\{ b \sum_{\tau_h \leq t} \frac{1}{a_h} - \eta c_g(t) \right\}. \quad (90)$$

The sum includes the initial choices. After a type exhausts, its recorded entry count stays fixed; this expression is used only for surviving types. Since $r_g(t) \leq q$ and $\sum_g r_g(t) = mqu$, weighted Jensen gives

$$\begin{aligned} \sum_g \tilde{M}_g(t) &\geq \frac{1}{q} \sum_g r_g(t) \exp \left\{ b \sum_{\tau_h \leq t} \frac{1}{a_h} - \eta c_g(t) \right\} \\ &\geq mu \exp \left\{ b \sum_{\tau_h \leq t} \frac{1}{a_h} - \frac{\eta \mathcal{W}(t)}{mqu} \right\}. \end{aligned} \quad (91)$$

Taking logarithms and expectations and using $\mathbb{E} \sum_g \tilde{M}_g(t) \leq m$ yields

$$b \mathbb{E} \sum_{\tau_h \leq t} \frac{1}{a_h} \leq \frac{\eta \mathbb{E} \mathcal{W}(t)}{mqu} - \log u. \quad (92)$$

Multiply by u and integrate over $t \in [0, 1]$. By (88), the left-hand side integrates to $bq \int_0^1 f_q(t) dt$. The supply identity (40) and $\int_0^1 u(-\log u) du = 1/4$ give

$$bq \int_0^1 f_q(t) dt \leq \frac{\eta q}{2} + \frac{1}{4}, \quad (93)$$

which is (42). Together with Step 1 this proves (43).

Step 3: choose the two parameters. Set $\eta := q^{-1/2}$, write $x := \eta \leq 1$, and put $C_q := (x/2 + x^2/4)/(1 - e^{-x})$. The elementary inequality $(1 - e^{-x})^{-1} \leq x^{-1} + 1/2 + x/12$ gives

$$2C_q \leq 1 + x + x^2/3 + x^3/24 \leq (1 + 0.55x)^2. \quad (94)$$

Minimizing $\log m/(q\theta) + C_q\theta$ over $\theta > 0$ in (43) yields

$$f_q(1) - 1 \leq \sqrt{2 \log m/q} (1 + 0.55/\sqrt{q}). \quad (95)$$

Lemma 3 proves (44). For completeness, the reciprocal inequality follows after multiplication from $(x^2 - 6x + 12)e^x \geq x^2 + 6x + 12$: the coefficients through degree four cancel, and the coefficient of x^k for $k \geq 5$ is $(k-3)(k-4)/k! \geq 0$. $\square$

B Proofs for the comparison with Random Priority

B.1 RP ranks and quotas

LEMMA 6. *For every $m \geq 2$ and $q \geq 1$,*

$$\max \left\{ 1, \frac{\mathbb{E} \text{Rank}(\text{RP})|_{q=1}}{q} \right\} \leq \mathbb{E} \text{Rank}(\text{RP}) \leq 1 + \sqrt{\frac{2 \log m}{q}} + \frac{\log m}{q}. \quad (96)$$

where $\mathbb{E} \text{Rank}(\text{RP})|_{q=1}$ denotes the expected average rank with one copy of each type. For every fixed m , $\mathbb{E} \text{Rank}(\text{RP})$ is non-increasing in the integer quota q .

PROOF. We use one auxiliary clock representation for both bounds and monotonicity. At each step, the next agent's ranking is independent of the earlier choices. If a types remain available, her favourite among them is uniformly distributed, and its expected rank is $(m+1)/(a+1)$. We can therefore reproduce the distribution of RP's sequence of chosen types by giving each type an independent rate-one Poisson clock: the next ring selects uniformly among the active clocks, uses one copy of that type, and stops its clock if this was the last copy. Only the order of the rings matters for the allocation.

Let G_j be the time of the q th ring of clock j . These times are independent $\text{Gamma}(q, 1)$ variables, each with mean q and variance q . Writing F_q for their distribution function, the number of active clocks is $A(t) = |\{j : G_j > t\}| \sim \text{Bin}(m, 1 - F_q(t))$. When $A(t) = a$, allocations occur at rate a , each with expected rank $(m+1)/(a+1)$. Integrating the expected sum of ranks over this auxiliary time and evaluating the binomial expectation gives

$$\begin{aligned} \mathbb{E} \text{Rank}(\text{RP}) &= \frac{m+1}{mq} \int_0^\infty \mathbb{E} \left[\frac{A(t)}{A(t)+1} \right] dt \\ &= \frac{1}{mq} \sum_{k=1}^m \mathbb{E} \max_{j \leq k} G_j. \end{aligned} \quad (97)$$

Indeed, the binomial identity is $(m+1)\mathbb{E}[A(t)/(A(t)+1)] = \sum_{k=1}^m (1 - F_q(t)^k)$, and integrating each term gives the corresponding expected maximum. Writing each G_j as a sum of q independent exponential waiting times, it is at least its first waiting time. Applying this coupling to every maximum in (97) gives the lower bound

$\mathbb{E} \text{Rank}(\text{RP}) \geq \mathbb{E} \text{Rank}(\text{RP})|_{q=1}/q$, established by Nikzad [2022, Proposition 5.2, proof]. Every rank is also at least one.

For the upper bound, define $\mathcal{I}(r) := r - 1 - \log r$ for $r \geq 1$, and let $r_m > 1$ be the unique solution of $\mathcal{I}(r_m) = \log m/q$. The function $\mathcal{I}$ is strictly increasing above one. For $0 < \lambda < 1$, the gamma moment-generating function and Jensen's inequality give

$$\mathbb{E} \text{Rank}(\text{RP}) \leq \frac{1}{q} \mathbb{E} \max_{j \leq m} G_j \leq \frac{\log m/q - \log(1 - \lambda)}{\lambda}. \quad (98)$$

Choosing $\lambda := 1 - 1/r_m$ makes the last expression equal to r_m . To obtain the stated explicit bound, put $x := \sqrt{2 \log m/q}$. Since $e^x \geq 1 + x + x^2/2$,

$$\mathcal{I}(1 + x + x^2/2) = x + x^2/2 - \log(1 + x + x^2/2) \geq x^2/2 = \log m/q.$$

Monotonicity of $\mathcal{I}$ therefore gives $r_m \leq 1 + \sqrt{2 \log m/q} + \log m/q$. Since $\mathbb{E} \text{Rank}(\text{RP})|_{q=1} = \log m + O(1)$ by Lemma 1, these bounds give $\mathbb{E} \text{Rank}(\text{RP}) \sim (\log m)/q$ when $q = o(\log m)$, and $\mathbb{E} \text{Rank}(\text{RP}) \rightarrow 1$ whenever $(\log m)/q \rightarrow 0$. Finally, G_j/q is a mean of q independent exponentials. A vector of $(q + 1)$ -sample means is the average of its leave-one-out vectors of q -sample means. Convexity of the maximum and (97) therefore give monotonicity in q . $\square$

B.2 Proof of Theorem 2

PROOF. We consider the three regimes in turn. If $q = o(\log m)$, set $a := \log m/q \rightarrow \infty$. By Lemma 6, $\mathbb{E} \text{Rank}(\text{RP}) \sim a$, whereas (44) gives $\mathbb{E} \text{Rank}(\text{PS}) \leq 1 + 1.55\sqrt{2a} = o(a)$. This proves (13).

Next suppose $q/\log m \rightarrow c \in (0, \infty)$. The root r_c has a direct interpretation. For large q , the probability that a clock's q th ring occurs after time qr has leading exponential term $e^{-q\mathcal{I}(r)}$. Among m clocks, the transition occurs around $me^{-q\mathcal{I}(r)} = 1$, or $\mathcal{I}(r) = \log m/q \rightarrow 1/c$. The following bounds make this intuition precise.

The bound (98), with $\lambda = 1 - 1/r_m$ as above, gives $\mathbb{E} \text{Rank}(\text{RP}) \leq r_m \rightarrow r_c$.

For a lower bound, fix $y > 1$. Integrating the $\text{Gamma}(q, 1)$ density over $[qy, qy+1]$ and applying Stirling's formula gives

$$\mathbb{P}(G_1 \geq qy) \geq \exp\{-q\mathcal{I}(y) - O(\log q)\} \quad (q \rightarrow \infty). \quad (99)$$

Fix $\delta \in (0, 1)$ and put $k := \lfloor \delta m \rfloor$. For $1 < y < r_c$, we have $c\mathcal{I}(y) < 1$, and hence

$$\log\{k\mathbb{P}(G_1 \geq qy)\} \geq \log m - q\mathcal{I}(y) - O(\log q) + O(1) \longrightarrow \infty. \quad (100)$$

Independence gives $\mathbb{P}(\max_{j \leq k} G_j \geq qy) \rightarrow 1$. The summands in (97) are non-decreasing in k , so

$$\liminf_{m \rightarrow \infty} \mathbb{E} \text{Rank}(\text{RP}) \geq (1 - \delta)y. \quad (101)$$

Letting $y \uparrow r_c$ and $\delta \downarrow 0$ proves $\mathbb{E} \text{Rank}(\text{RP}) \rightarrow r_c$. Since $q \rightarrow \infty$, (44) gives $\limsup_{m \rightarrow \infty} \mathbb{E} \text{Rank}(\text{PS}) \leq 1 + \sqrt{2/c}$. For every $s > 0$, $\mathcal{I}(1+s) = s - \log(1+s) < s^2/2$: the difference $s^2/2 - s + \log(1+s)$ is zero at zero and has derivative $s^2/(1+s) > 0$. Taking $s = \sqrt{2/c}$ and using strict monotonicity of $\mathcal{I}$ on $(1, \infty)$ proves (15). In particular,

$$\liminf_{m \rightarrow \infty} (\mathbb{E} \text{Rank}(\text{RP}) - \mathbb{E} \text{Rank}(\text{PS})) \geq r_c - 1 - \sqrt{2/c} > 0. \quad (102)$$

Finally, if $q/\log m \rightarrow \infty$, (44) and (96), together with the lower bound one, give (16). This proves sufficiency in (17). For necessity, if $q/\log m$ does not tend to infinity, there is a bounded subsequence and hence a further subsequence converging to some $c \in [0, \infty)$. If $c = 0$, the rank gap diverges by (13); if $c > 0$, its lower limit is strictly positive by (15). Thus the absolute difference of expected ranks cannot converge to zero. $\square$

C Proofs of the extensions

The extensions use Lemma 4 and the same supply calculation. If a set T survives throughout a phase $[a, b]$, each of its types has capacity at most Q , and its total number of consumers is at least $dn|T|/A(t)$, then

$$dn|T| \int_a^b \frac{1}{A(t)} dt \leq \int_a^b \sum_{g \in T} c_g(t) dt \leq Q|T|, \quad \int_a^b \frac{m}{A(t)} dt \leq \frac{mQ}{dn}. \quad (103)$$

Thus only the ratio of supply per type to consumers per type changes the phase bound.

C.1 Unequal quotas

PROOF OF PROPOSITION 3. Write $\bar{q} := n/m$ and $q_{\max} := \max_g q_g \leq K\bar{q}$. Apply Lemma 4 with $D = \kappa = 1$. Its draw and consumer counts do not use the individual capacities, and its failure bound holds for every real $\bar{q} \geq 1$.

Use the phases from Proposition 2. In (103), take $Q = q_{\max}$ and $d = \eta/8$. Each phase contributes at most $8K/\eta$ to $\int_0^1 m/A(t) dt$. The phase thresholds and initial contribution are unchanged. If at most two types remain, their supply is at most $2q_{\max}$, so the remaining contribution is at most $2q_{\max}/\bar{q} \leq 2K$. The draw bound (60) and the pathwise rank comparison therefore give the claimed high-probability bound. Since $\text{Rank}(\text{PS}) \leq m \leq n$, the exceptional event contributes $O(n^{-2}) + ne^{-cn}$ to the expectation. $\square$

C.2 Unequal numbers of agents and goods

Use the assignment conventions of Section 5.2. Until time $\tau := \min\{1, m/n\}$, all n agents are consuming. The sampling construction and the pathwise bound on rank by draw count therefore apply up to that time.

C.2.1 More goods than agents.

PROOF OF THEOREM 3. Apply Lemma 4 with $D = B$ and $\kappa = 1$, and denote its constants by $C_B, \Lambda_B, \eta_B, \alpha_B$. On its good event,

$$\text{Rank}(\text{PS}) \leq C_B \int_0^1 \frac{m}{A(t)} dt. \quad (104)$$

Define $b_B(s) := \lceil m \exp(-\alpha_B m/s) \rceil$. Start the phase construction when $A(t)$ first falls below m/Λ_B ; the preceding part of the integral is at most Λ_B . If this never happens before time one, the bound follows immediately. Otherwise, at the start t_j of a phase put $\ell_j := A(t_j)$ and end the phase when $A(t)$ first falls to at most $2b_B(\ell_j)$ or time one is reached, whichever comes first.

For any positive-length phase with $\ell_j > 2$, the types surviving just before its endpoint form a set T_j with more than $2b_B(\ell_j)$ elements. Since $b_B(A(t)) \leq b_B(\ell_j)$ throughout the phase, (62) applies to T_j throughout. Taking $Q = 1$ and $d = \eta_B/8$ in (103) bounds the phase contribution by $8B/\eta_B$.

The excess supply shortens the number of phases. If a threshold is reached before time one, write $\lambda_j := m/\ell_j$. Whenever $me^{-\alpha_B \lambda_j} \geq 1$,

$$\lambda_{j+1} \geq \frac{1}{4}e^{\alpha_B \lambda_j}.$$

Choose the initial threshold larger, if necessary, so that $x_j := \alpha_B \lambda_j/2$ starts above $\max\{e, \log(8/\alpha_B)\}$. Then

$$x_{j+1} \geq \frac{\alpha_B}{8}e^{2x_j} \geq e^{x_j},$$

so these rescaled thresholds grow at least by repeated exponentiation.

At time t , total remaining supply is $m - nt$, and no type has more than one unit. Consequently,

$$A(t) \geq \Delta + n(1 - t) > \Delta \quad (t < 1). \quad (105)$$

If $\Delta \geq 4$, once $me^{-\alpha_B \lambda_j} \leq \Delta/4$, the next threshold is at most Δ , so (105) forces that phase to end at time one. If $\Delta \leq 3$, once $me^{-\alpha_B \lambda_j} < 1$, the next threshold is two. Reaching it leaves supply at most two and hence contributes at most $2B$ to the integral.

Thus the phase construction stops after $\alpha_B \lambda_j$ exceeds $\log(4m/\max\{\Delta, 4\})$ by a fixed constant. Since

$$\frac{m}{\max\{\Delta, 4\}} \leq 1 + \frac{n}{\Delta_+},$$

repeated exponentiation reaches this level in $O_B(\log^*(1 + n/\Delta_+))$ phases. This proves (48) on the good event. The exceptional contribution is $o(1)$ because rank is at most $m \leq Bn$.

For RP, the k th agent faces $s = m - k + 1$ goods. This set is independent of her ranking, so her expected best rank is $(m + 1)/(s + 1)$. Averaging over $k = 1, \dots, n$ gives (49). If $\Delta = o(n)$, this expression is of order $\log(1 + n/\Delta_+)$, which dominates its iterated logarithm. If Δ/n does not tend to zero, RP's expected rank is bounded along a subsequence while PS's is at least one, so their ratio cannot tend to zero. Harmonic-number asymptotics give the final limit. $\square$

C.2.2 More agents than goods.

PROOF OF THEOREM 4. PS exhausts all goods at time m/n . Multiplying time and every capacity by n/m turns this market into a replica market with m types, capacity $q = n/m$ for each type, and n agents eating on $[0, 1]$. All allocated shares and the total allocated mass are multiplied by the same factor, so average rank per allocated unit equals the replica market's average rank.

The expectation bounds in (6) and (44) also hold for real capacities $q \geq 1$: their proofs use $q \geq 1$, but do not require integrality. Substitute $q = n/m$ and use $0.55\sqrt{2 \log m} \leq \log m$ for $m \geq 2$ to obtain (53); the case $m = 1$ is immediate.

Under RP, only the first m agents in the priority order receive goods. Their preferences remain independent and uniform, so RP's expected average rank per allocated unit equals that of the balanced market with m agents and m goods. This gives (54), and the asymptotic conclusions follow. $\square$

C.3 Correlated preferences

Use the one-copy model in Definition 1. Conditional on the common weights, give each agent an independent stream in which label g has probability $p_g := w_g/W(G)$, where $W(S) := \sum_{g \in S} w_g$. The first unseen label is chosen proportionally to its weight, so first appearances generate exactly the specified Plackett–Luce ranking. Skipping exhausted types therefore produces PS as in the uniform construction. Normalize the weights so that $1 \leq w_g \leq \kappa$; then $1/(\kappa m) \leq p_g \leq \kappa/m$.

PROOF OF THEOREM 5. Conditional on every admissible weight vector, Lemma 4 applies with $D = 1$ and the stated κ . The pathwise rank comparison gives

$$\text{Rank(PS)} \leq C_\kappa \int_0^1 \frac{m}{A(t)} dt$$

outside its exceptional event. Use the phases of Proposition 2, with the constants from that lemma. By (103), each phase contributes at most $8\kappa/\eta$, and the thresholds grow at least by repeated exponentiation. The resulting $O_\kappa(\log^* m)$ bound and the exceptional-event estimates are uniform in the weights, so the same expectation bound holds after averaging over them.

For RP, condition on the weights and the set S of s goods remaining when an agent chooses. Her ranking is independent of S and can also be generated by ordering independent exponential clocks with rates w_g : each next clock rings first with probability proportional to its rate. An outside good $g \notin S$ precedes every good in S with probability $w_g/(W(S) + w_g)$. Counting these outside goods gives the exact identity

$$\mathbb{E}[\rho_i(S) \mid S, w] = 1 + \sum_{g \notin S} \frac{w_g}{W(S) + w_g}. \quad (106)$$

Since $1 \leq w_g \leq \kappa$,

$$\frac{m+1}{\kappa(s+1)} \leq \mathbb{E}[\rho_i(S) \mid S, w] \leq \kappa \frac{m+1}{s+1}.$$

Average over the positions $s = m, m-1, \dots, 1$ and then over the weights. The resulting expected rank is $\Theta_\kappa(H_m) = \Theta_\kappa(\log m)$. $\square$

PROPOSITION 4. *Suppose one strict order is drawn uniformly and every agent has that same order. Each agent's marginal preference is uniform, but*

$$\text{Rank(PS)} = \text{Rank(RP)} = \frac{m+1}{2} \quad (107)$$

for every m .

PROOF. Every feasible complete assignment allocates exactly one unit of each good. With a common ranking, its average rank is therefore $m^{-1} \sum_{r=1}^m r = (m+1)/2$, under either mechanism. $\square$